

Introduction to Relation

Lecture-5

Relations



Relations and Their Properties
Representing Relations
Equivalence Relations
Partial Orderings

Relations and Their Properties

Let A and B be two arbitrary sets, A binary relation on (or from) A to B is a subset “ f ” of $A \times B$.

Example:

- Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$
- $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B .

Relations and Their Properties
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Relations and Their Properties

If $A = B$, “ f ” is called relation on A , if $(x,y) \in f$, we often write this as $x f y$ and say that x is related to y through f . (xRy)

Note: Since $\emptyset \subseteq A \times B$, $\emptyset$ is a relation from A to B , also, as $A \times B \subseteq A \times B$, $A \times B$ is a relation from A to B .

Domain, Range and Inverse

Domain of $f =$

$$D_f = \{x \in A: (x, y) \in f \text{ for some } y \in B\}$$

Range of $f =$

$$R_f = \{y \in B: (x, y) \in f \text{ for some } x \in A\}$$

Inverse of $f =$

$$R_f = \{(y, x): (x, y) \in f\}$$

Domain, Range and Inverse

Example: Let $A = B = \{1, 2, 3, 4, 5\}$ and if

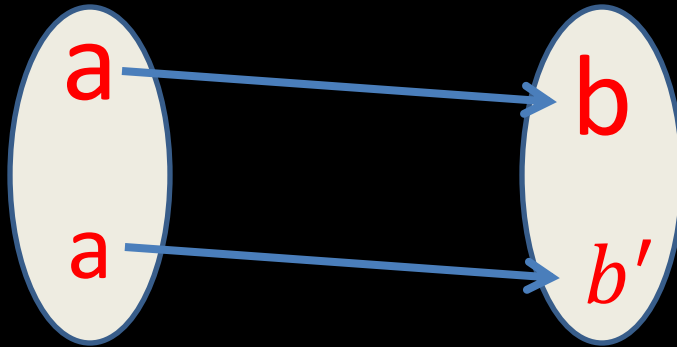
$$(i) f = \{(x, y) \in A \times B : x = y\}$$

$$(ii) f = \{(x, y) \in A \times B : x < y\}$$

Find f , Domain, Range

One-one , one –many, many –one Relation

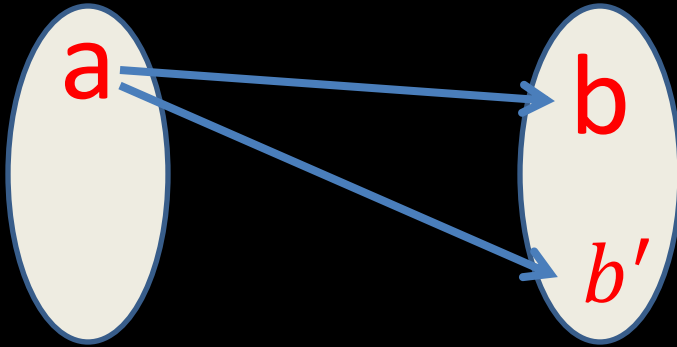
(i) one-one



$$(a, b) \in f, (a, b') \in f \Rightarrow b = b'$$

One-one , one –many, many –one Relation

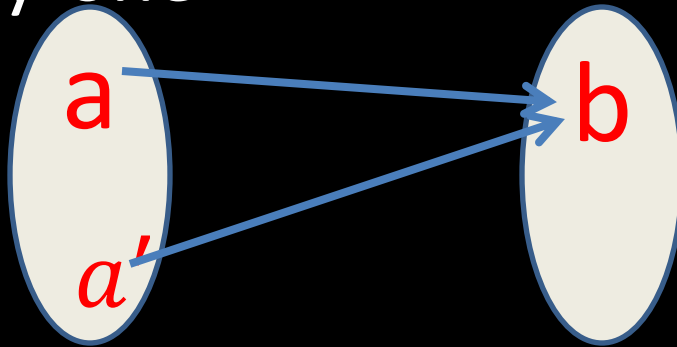
(i) one-many



$$(a, b) \in f, (a, b') \in f \Rightarrow b \neq b'$$

One-one , one –many, many –one Relation

(i) many-one



$$(a, b) \in f, (a', b) \in f \Rightarrow a \neq a'$$

Reflexive, symmetric, transitive

*A relation f on a set A is called “**Reflexive**” if $a f a$ that is $(a, a) \in f$ for every $a \in A$.*

*A relation f on a set A is called “**Symmetric**” if $(a, b) \in f \Rightarrow (b, a) \in f$ or equivalently $f = f^{-1}$.*

*A relation f on a set A is called “**Transitive**” if $(a, b) \in f$ and $(b, c) \in f \Rightarrow (a, c) \in f$.*

Relations and Their Properties

Reflexive, symmetric,
transitive

Parallel of two lines is reflexive.

Perpendicular of two lines is symmetric.

Parallel of three lines is “Transitive ”

Relations and Their Properties

Reflexive, symmetric,
transitive

Fatherhood relation is neither reflexive nor symmetric, nor transitive"

Ancestor relation satisfies only **transitive** condition.

Spouse relation satisfies only **symmetric** condition.

Relations and Their Properties

Anti symmetric , equivalence relation , Partial Ordering

A relation f on a set A is called “*Anti symmetric*” if $(a, b) \in f$ and $(b, a) \in f$ then $a = b$.

A relation f on a set A is called “*equivalence relation*” if f satisfies the (i) reflexive (ii) symmetric (iii) Transitive

A relation f on a set A is called “*Partial Ordering relation*” if f satisfies the (i) reflexive (ii) anti-symmetric (iii) Transitive

Relations and Their Properties

*Anti symmetric , equivalence
relation , Partial Ordering*

Subset relation of two set satisfies the “Anti symmetric ” relation only.

Division relation is not “equivalence relation ”

Subset relation is “Partial Ordering relation ”

*A relation f on a set A is called “**Total Ordering**” if it is partial ordering and either $(a, b) \in f$ or $(b, a) \in f$ for $a, b \in A$.*

Relations and Their Properties

Combining Relations.

Given two relations R_1 and R_2 , we can combine them using basic set operations to form new relations such as $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, and $R_2 - R_1$. $R_1 \oplus R_2 = R_1 \cup R_2 - R_1 \cap R_2$

Example: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$. The relations $R_1 = \{(1,1),(2,2),(3,3)\}$ and $R_2 = \{(1,1),(1,2),(1,3),(1,4)\}$ can be combined using basic set operations to form new relations:

$$R_1 \cup R_2 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(3,3)\}$$

$$R_1 \cap R_2 = \{(1,1)\}$$

$$R_1 - R_2 = \{(2,2),(3,3)\}$$

Relations and Their Properties

Combining Relations.

Let R_1 be the “Less than” relation on the set of real numbers and R_2 be the “greater than” relation on the set of real numbers, that is $R_1 = \{(x, y): x < y\}$ and $R_2 = \{(x, y): x > y\}$ then what is $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, and $R_2 - R_1$, $R_1 \oplus R_2$

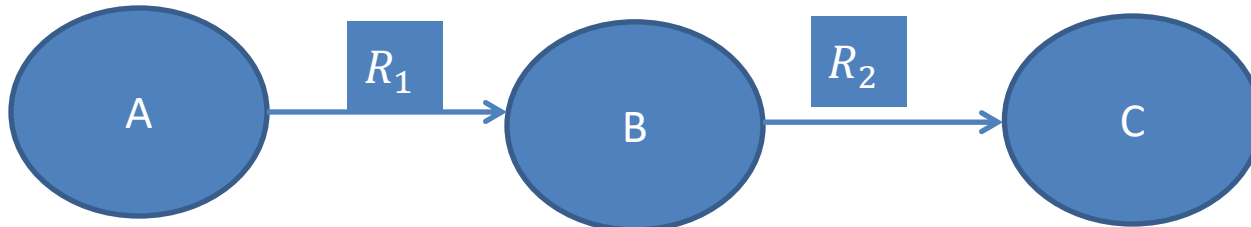
Definition: Suppose

R_1 is a relation from a set A to a set B .

R_2 is a relation from B to a set C .

Then the *composition* (or *composite*) of R_2 with R_1 , is a relation from A to C where

if (a,b) is a member of R_1 and (b,c) is a member of R_2 , then (a,c) is a member of $R_2 \circ R_1$.



*What is the composite of the relation $R_1 \circ R_2$, if
 $R_1 = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$ and
 $R_2 = \{(1,0), (2,0), (3,1), (3,2), (4,1)\}$*

Definition: Let R be a binary relation on A . Then the powers R^n of the relation R can be defined recursively by:

Basis Step: $R^1 = R$

Recursive Step: $R^{n+1} = R^n \circ R$

Let $R = \{(1,1), (1,4), (2,3), (3,1), (3,4)\}$ find the powers R^n , $n=1,2,3,4$

Relations and Their Properties

Representing Relations

- Representing Relations using Matrices
- Representing Relations using Digraphs

- Representing Relations using Matrices

- Representing Relations using Matrices
- A relation between finite sets can be represented using a zero-one matrix.

The relation R is represented by the $m \times n$ matrix $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R, \\ 0 & \text{if } (a_i, b_j) \notin R. \end{cases}$$

The matrix representing R has a 1 as its (i,j) entry when a_i is related to b_j and a 0 if a_i is not related to b_j .

Examples of Representing Relations Using Matrices

Example 1: Suppose that $A = \{1,2,3\}$ and $B = \{1,2\}$. Let R be the relation from A to B containing (a,b) if $a \in A$, $b \in B$, and $a > b$. What is the matrix representing R

Solution: Because $R = \{(2,1), (3,1), (3,2)\}$, the matrix is

Relations and Their Properties

Representing Relations

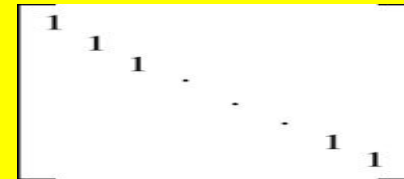
$$M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. Which ordered pairs are in the relation R represented by the matrix

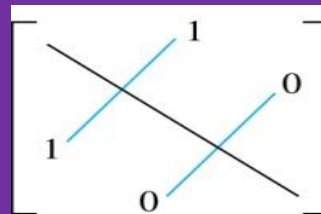
$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix} ?$$

Solution: Because R consists of those ordered pairs (a_i, b_j) with $m_{ij} = 1$, it follows that: $R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3), (a_3, b_5)\}$.

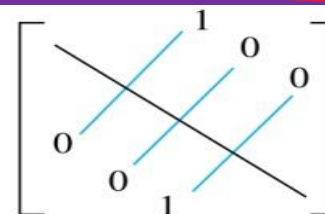
If R is a reflexive relation, all the elements on the main diagonal of M_R are equal to 1.



R is a symmetric relation, if and only if $m_{ij} = 1$ whenever $m_{ji} = 1$. R is an antisymmetric relation, if and only if $m_{ij} = 0$ or $m_{ji} = 0$ when $i \neq j$.



(a) Symmetric



(b) Antisymmetric

Example : Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

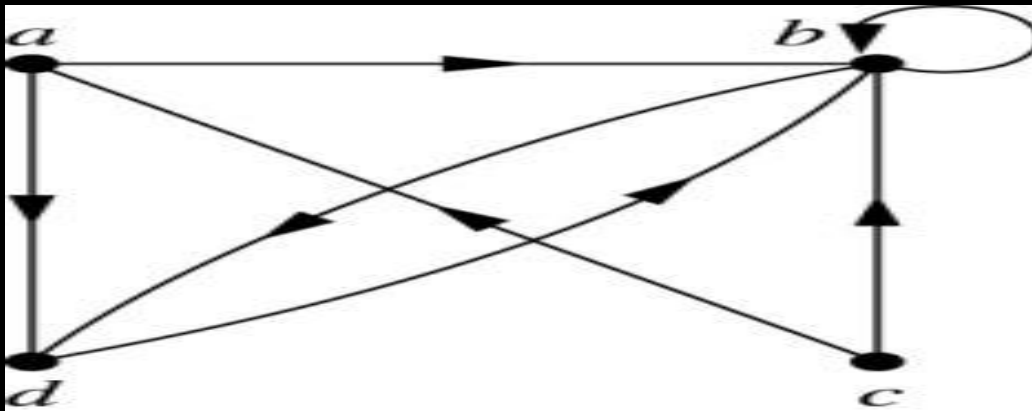
Is R reflexive, symmetric, and/or antisymmetric?

Solution: Because all the diagonal elements are equal to 1, R is reflexive. Because M_R is symmetric, R is symmetric and not antisymmetric because both $m_{1,2}$ and $m_{2,1}$ are 1.

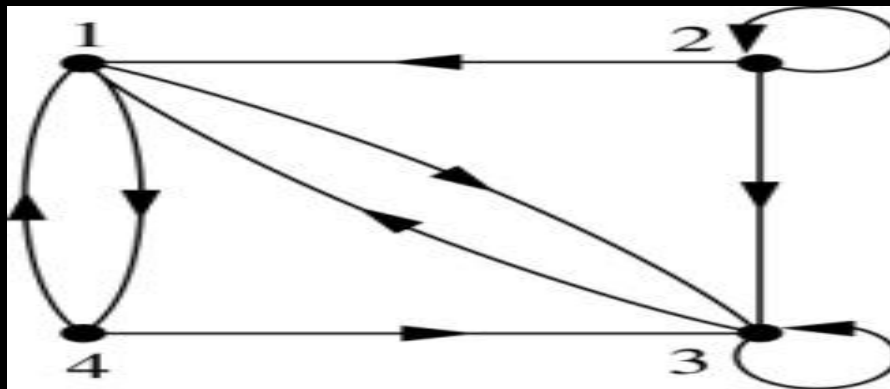
Definition: A *directed graph*, or *digraph*, consists of a set V of *vertices* (or *nodes*) together with a set E of ordered pairs of elements of V called *edges* (or *arcs*). The vertex a is called the *initial vertex* of the edge (a,b) , and the vertex b is called the *terminal vertex* of this edge.

An edge of the form (a,a) is called a *loop*.

Example : A drawing of the directed graph with vertices a , b , c , and d , and edges (a, b) , (a, d) , (b, b) , (b, d) , (c, a) , (c, b) , and (d, b) is shown here.



Example : What are the ordered pairs in the relation represented by this directed graph?



Solution: The ordered pairs in the relation are
(1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (3, 1), (3, 3), (4, 1), and (4, 3)

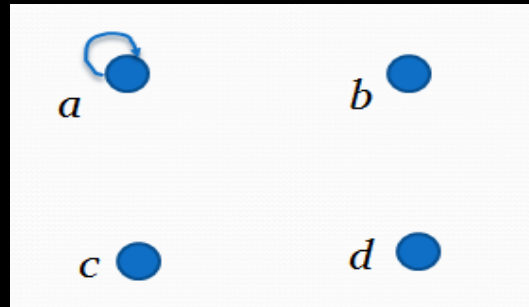
Reflexivity: A loop must be present at all vertices in the graph.

Symmetry: If (x,y) is an edge, then so is (y,x) .

Anti symmetry: If (x,y) with $x \neq y$ is an edge, then (y,x) is not an edge.

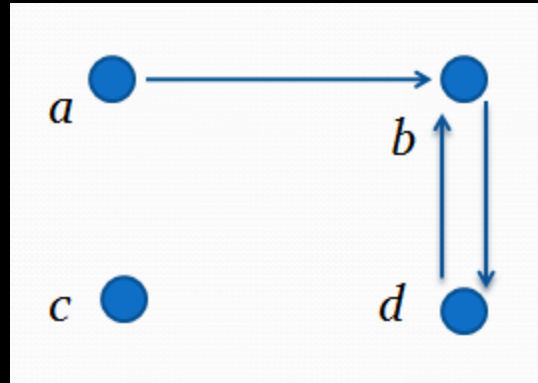
Transitivity: If (x,y) and (y,z) are edges, then so is (x,z) .

Determining Properties of a Relation from its Digraph



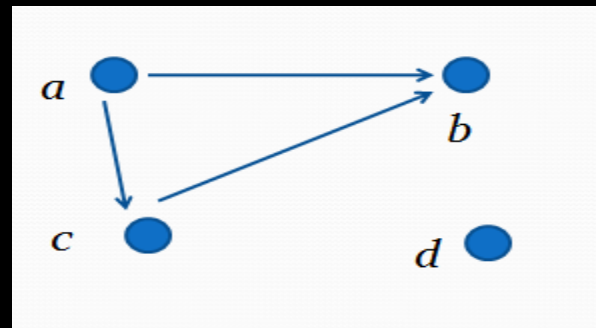
- **Solution:** *Reflexive?* No, not every vertex has a loop
- *Symmetric?* Yes (trivially), there is no edge from one vertex to another
- *Antisymmetric?* Yes (trivially), there is no edge from one vertex to another
- *Transitive?* Yes, (trivially) since there is no edge from one vertex to another

Determining Properties of a Relation from its Digraph



- *Reflexive?* No, there are no loops
- *Symmetric?* No, there is an edge from a to b , but not from b to a
- *Antisymmetric?* No, there is an edge from d to b and b to d
- *Transitive?* No, there are edges from a to c and from c to b , but there is no edge from a to d

Determining Properties of a Relation from its Digraph



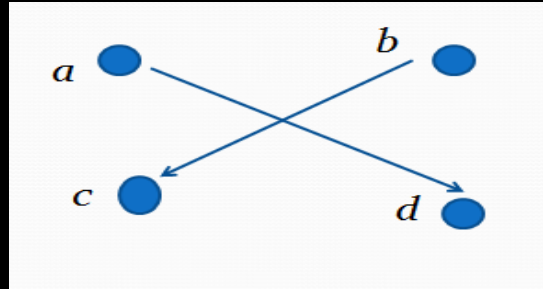
Reflexive? No, there are no loops

Symmetric? No, for example, there is no edge from c to a

Anti symmetric? Yes, whenever there is an edge from one vertex to another, there is not one going back

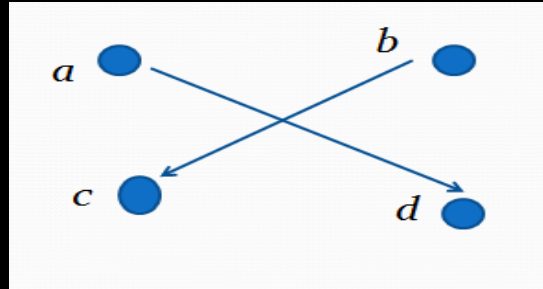
Transitive? Yes, there is no edge from a to b

Determining Properties of a Relation from its Digraph



- *Reflexive?* No, there are no loops
- *Symmetric?* No, for example, there is no edge from d to a
- *Antisymmetric?* Yes, whenever there is an edge from one vertex to another, there is not one going back
- *Transitive?* Yes (trivially), there are no two edges where the first edge ends at the vertex where the second edge begins

Determining Properties of a Relation from its Digraph



- *Reflexive?* No, there are no loops
- *Symmetric?* No, for example, there is no edge from d to a
- *Antisymmetric?* Yes, whenever there is an edge from one vertex to another, there is not one going back
- *Transitive?* Yes (trivially), there are no two edges where the first edge ends at the vertex where the second edge begins